Hearing Valley:

Optimizing Low Frequency Sound Pressure Level by Simulation in Hearing Valley Room Geometry

Pilot Project 2025–2026

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Abstract

Hearing loss is a disability that has an effect on hundreds of millions of people. People who suffer from hearing loss have an elevated risk of dementia or Alzheimer's. Hearing loss is usually treated with hearing aid. Hearing aids in the market today are designed to only amplify and reduce noise of the sound. There is a need for innovating new hearing aid technology, which can produce spatial audio to the wearer.

Hearing Valley is a multidisciplinary research, development, and innovation (RDI) infrastructure built for audiology research and signal processing. It was founded to bring together researchers, Kuopio University Hospital (KUH), and local businesses to innovate the future of hearing rehabilitation. Hearing Valley has physical audio room built to test hearing, or audiology related innovations, with loudspeaker system.

The Hearing Valley room has a well-designed acoustical system. That is, it has highly absorbent walls and ceiling. These acoustical conditions were modelled in COMSOL Multiphysics for Hearing Valley room geometry. The parameters needed for creating acoustic impedance models were received from the manufacturers of the walls and the ceiling. Acoustical impedance model for ceiling and walls were separately derived by extrapolating and interpolating absorption coefficient data obtained from the manufacturers. The room geometry was also built in COMSOL.

The project task was to optimize the frequency for the listener response at low frequencies using COMSOL simulations within the Hearing Valley room geometry. The task was done by solving Helmholtz equation inside Hearing Valley room geometry with finite element methods implemented in COMSOL Multiphysics. The optimization was done by optimizing the listener's location inside the room geometry. The aim was to gain as flat sound pressure level as possible.

Validation measurements were also done in Hearing Valley physical room. The measurement set-up was done with two subwoofers and a single, omnidirectional measurement condenser microphone placed to the fixed location of the listener. These measurements were analyzed and compared with the simulation results. Since the listener's location is fixed in the physical room, the measurement results could only be compared to the original location of the listener from the simulations.

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Schedule

Table 1: Project schedule. Total hours used for this project was 148 h.

Topic	December 2025	January 2026	February 2026
COMSOL	20 h	30 h	30 h
Pressure Acoustics	10 h	5 h	5 h
Meetings	3 h	3 h	2 h
Writing report	10 h	10 h	20 h
Total:	**43 h**	**48 h**	**57 h**

1 Introduction

Disabling hearing loss is a disability that directly affects around 5 % of the global population [1]. The definition of disabling hearing loss is 35 dB loss in the better hearing ear, according to World Health Organization (WHO). If a person has a hearing threshold of 20 dB, they have hearing loss [1]. Based on this information, hearing loss has a huge effect on the quality of life of individuals.

Hearing loss has a direct effect on person's cognitive capabilities and communication. It raises the risk of dementia and Alzheimer's [2, 3]. These factors justify to keep studying hearing loss, and develop hearing aids.

Hearing aids are sophisticated, computerized electroacoustic devices. Modern hearing aid technology consists of a microphone, an electronic amplification circuit, and a miniature loudspeaker to amplify the sound; with a battery powering the electronic parts [4]. The microphone capturing the sound can be directed to the foreground of the person wearing the hearing aid. This amplifying circuit doesn't create spatial audio. Modern hearing aids can, for example, amplify the signal from the microphone and reduce the background noise [4].

There is some research and development done towards machine produced spatial audio, such as Kardome spatial hearing AI products [5] and Aalto University research [6]. The spatial sound hearing aid is studied, but the technology is too energy consuming. It cannot be applied to such small, wearable device as hearing aid [7].

Hearing Valley is a multidisciplinary research, development, and innovative (RDI) infrastructure built for audiology research. It was founded in tandem with University of Eastern Finland (UEF) and Kuopio University Hospital (KUH) for hearing research and future innovations for hearing rehabilitation. It enables local companies to expand their operations towards

innovating new technology into the Finnish market [8]. New innovations in hearing rehabilitations is needed due to the increase of noise exposure of individuals [8].

This project work is a part of the Hearing Valley project. The objective of this project is to optimize sound pressure level (SPL) for low frequency acoustics by simulations in COMSOL Multiphysics, and validate the results with measurements. The goal is to achieve a flat SPL response. Optimization is done by finding the optimal location for the listener. This is done to study the stability of the frequency response for the listener's ear positions, because the listener's location is fixed inside the physical Hearing Valley room.

In Section 2, the essential theory of pressure acoustics is introduced. It involves Helmholtz equation in pressure acoustics, boundary conditions for acoustic system, and sound pressure level. Section 3 goes through the methods implemented in the simulations and the validation measurements of the simulation. Section 4 is about the results and findings of the study, and in Section 5, the results are discussed. Section 6 concludes the findings of this study.

2 Theory

Wave equation is important in modeling physical acoustics. With wave equation we can derive several important concepts and quantities related to acoustics. Homogenous wave equation can be written in format

$$\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \Delta p, \tag{2.1}$$

where $\frac{\partial^2 p}{\partial t^2}$ is the second derivative of pressure p with respect to time and Δp is the Laplace operation $\left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2}\right)$. The wave equation can be used in modeling acoustic waves, but in this project, we need to obtain the frequency content of the waves. That is, we are interested in the frequency response of the simulation geometry.

2.1 Helmholtz equation

Let's assume that the wave equation solution is time-harmonic, meaning that the solution can be written as

$$p(x,y,z,t) = p(x,y,z)e^{i\omega t}. \tag{2.2}$$

Now the equation (2.1) can be presented as

$$\Delta p e^{i\omega t} + \frac{1}{c^2}\omega^2 p e^{i\omega t} = 0, \tag{2.3}$$

where the second time derivative is

$$\frac{\partial^2 p}{\partial t^2} = -\omega^2 p e^{i\omega t}.$$

The time-harmonic term $e^{i\omega t}$ is a common factor for the equation and is always non-zero, so

$$\Delta p + \frac{\omega^2}{c^2} p = 0. \tag{2.4}$$

The equation (2.4) is known as Helmholtz equation. The solution for Helmholtz equation in acoustics is a function of acoustic pressure with respect to frequency. This function is called frequency response of an acoustic system.

Since Helmholtz equation is used for simulations, we need to derive the equation with some information of the acoustical medium and source term. The Helmholtz equation can be derived as

$$\frac{1}{\rho}\Delta p + \frac{k^2}{\rho} p = Q_m, \tag{2.5}$$

where ρ is the density of the medium, $k = \omega/c$, and Q_m is the source term.

2.2 Boundary Conditions

Boundary conditions (BC) are constraints for differential equations, which implies how the solution of the differential equation behaves on the boundaries of the system. BCs are a key step in creating a well-defined system with an unambiguous solution. Classical physics, such as mechanics, electromagnetism, and acoustics, requires well-defined systems for unambiguous and meaningful solutions.

Boundary conditions of acoustic systems define how the acoustic pressure wave behaves at the boundaries. Acoustic systems have 3 main BCs that can be used in defining the behavior of the pressure wave on the boundary; Neumann BC, Dirichlet BC, and Robin BC.

2.2.1 Neumann Boundary Condition

The Neumann boundary condition specifies the value of the normal derivative of the solved quantity on the boundary. That is

$$\frac{\partial p}{\partial \hat{\mathbf{n}}} = g, \text{ on } \partial\Omega, \tag{2.6}$$

where g is the source term that can be either constant or a function of some relevant quantity, Ω is the domain for the solution and $\partial\Omega$ is the boundary of the domain. In acoustics, if $g = 0$, this is known as the sound-hard BC. It is the BC for reflection without phase change of the pressure wave.

2.2.2 Dirichlet Boundary Condition

Dirichlet BC is generally defined as a quantity to be zero or constant in boundary. In the content of acoustics, Dirichlet BC denotes acoustic pressure to be constant, or a function of place, at boundaries. That is

$$p = a, \text{ on } \partial\Omega, \tag{2.7}$$

where a is the source term that can be either constant or function of some relevant quantity. If $a = 0$, Dirichlet BC is known as sound soft boundary in acoustics. It means that the pressure wave reflects out of phase.

2.2.3 Robin Boundary Condition

Neumann and Dirichlet BCs can be combined with different weights. The combination can be presented as a weighted sum of (2.6) and (2.7), that is

$$\alpha \frac{\partial p}{\partial \hat{\mathbf{n}}} + \beta p = y, \tag{2.8}$$

where α and β are weighting parameters and y is either a constant or function of some relevant quantity. The equation (2.8) is known as Robin BC.

2.3 Acoustic Impedance

Acoustic impedance is a medium's resistance to sound propagation. Normal specific impedance is defined in frequency domain as a relation of pressure and the normal incidence particle velocity, i.e.

$$z_n(\omega) = \frac{p(\omega)}{u(\omega) \cdot \hat{\mathbf{n}}}, \tag{2.9}$$

where both pressure p and particle velocity u are both functions of angular frequency $\omega = 2\pi f$ [9]. The equation (2.9) regards only the particle velocity component normal to the boundary. Sometimes, another version of specific impedance is used.

$$z_n = \rho c \frac{1+R}{1-R}, \tag{2.10}$$

where ρ is the density of the medium, c is the speed of sound in the medium, and $R = e^{i\phi}\sqrt{1-\alpha_n}$ is the pressure reflection coefficient. In pressure reflection coefficient, ϕ is the phase of the pressure wave and α_n is normal incidence absorption coefficient.

Acoustic impedance can be utilized in BCs. Let P be time-harmonic complex acoustic pressure

$$P = Ae^{i(\omega t - kr + \phi)}, \tag{2.11}$$

where t is time, r is the distance to the sound source, and ϕ is the phase of the wave. Let's examine the complex amplitude of the wave, p,

$$P = Ae^{i(\omega t - kr + \phi)} = Ae^{i(\phi - kr)}e^{i(\omega t)} =: pe^{i(\omega t)}. \tag{2.12}$$

Now Neumann BC (2.6) can be defined by differentiating (2.12) with respect to the normal of the boundary,

$$\frac{\partial p}{\partial \hat{\mathbf{n}}} = -i\omega\rho u_n. \tag{2.13}$$

With (2.9), the normal velocity $u_n = p/z_n$, the impedance BC becomes Robin BC; that is

$$\frac{\partial p}{\partial \hat{\mathbf{n}}} = -i\omega\rho\frac{1}{z_n}p \Leftrightarrow p + \frac{iz_n}{\omega\rho}\frac{\partial p}{\partial \hat{\mathbf{n}}} = 0. \tag{2.14}$$

The equation (2.14) is widely used as the impedance BC.

2.4 Sound Pressure Level

Sound pressure level (SPL) is an approach to express acoustic pressure with taking human hearing to account. SPL is a logarithmic measure of effective pressure related to a reference pressure. That is,

$$\mathrm{SPL} = 20\log_{10}\left(\frac{p}{p_0}\right) \tag{2.15}$$

where p is the amplitude of the acoustic pressure (always positive) and p_0 is the reference pressure. Air has the reference pressure $p_0 = 20\ \mu$Pa, which is used in this project.

3 Materials and Methods

3.1 COMSOL Multiphysics

COMSOL Multiphysics is a general-purpose simulation software, which can solve numerous physical problems with different modules [10]. It uses finite element methods to solve physical problems. COMSOL was used in this project for simulating the room geometry and the pressure acoustics inside the room. The geometry used in this project is presented in figure 8. The room geometry was provided by Ville Kallinen. Room configuration requires several parameters, such as BCs for the walls, ceiling, floor, listener figure, and speaker components (transducer, enclosure). All of the parameters relevant to pressure acoustics can be defined in COMSOL *Pressure Acoustics, Frequency Domain* module. The assignment was to solve and optimize SPL for low frequencies $(20 - 150$ Hz$)$ with changing the listener's location. The aim is for flat SPL response.

3.1.1 Pressure Acoustics, Frequency Domain

The pressure acoustics module of COMSOL numerically solves the Helmholtz equation derived in Section 2 (equation (2.5)). The numerical solver of partial differential equations (PDE) is finite element method (FEM). FEM solves PDEs numerically by discretizing the domain with smaller elements, and these elements build a mesh, in which the PDE is solved [11]. It is too time consuming to dive deeper into FEM theory for this project. Although it is relevant to know that COMSOL uses FEM to solve the physical problems.

COMSOL has different options for FEM solver. In COMSOL one can completely choose what kind of mesh is chosen for the solver. In this project, the mesh is physics-controlled, the element size was chosen as *Finer* for more accurate solution.

As noted in Section 2, differential equations need BCs to solve them unambiguously. Boundary conditions are set for this project as below.

Boundary Condition for Walls

Wall panels of the Hearing Valley room are Ewona Acustica Basic 50 mm acoustic panels. Impedance BC was used for setting the boundaries for the walls. The impedance was calculated with (2.10) and the normal incidence absorption coefficient α_n was set as a function of frequency. Ewona provided the data for the attenuation function, which describes the sound intensity decrease of wave propagation [12]. For simplicity, the attenuation coefficient values are used as absorption coefficient values for deriving the impedance of the boundaries. As one can see from Table 2, Ewona provides data only for 125 Hz and above. To solve α_n below 125 Hz, extrapolation methods are needed. Due to the lack of data, we used simple linear extrapolation to reach 20 Hz.

$$y = y_1 + (x - x_1)\frac{y_2 - y_1}{x_2 - x_2}, \tag{3.1}$$

where y is the wanted data value, y_1 and y_2 are known data points, x is wanted input value, and x_1 and x_2 are known input values. The extrapolation was done in MATLAB.

Table 2: Tabular values for attenuation coefficient of Ewona walls α_n provided by Ewona [12].

Frequencies (Hz)	125	250	500	1000	2000	4000
α_n (dB/m)	0.28	0.60	0.83	0.95	0.96	0.90

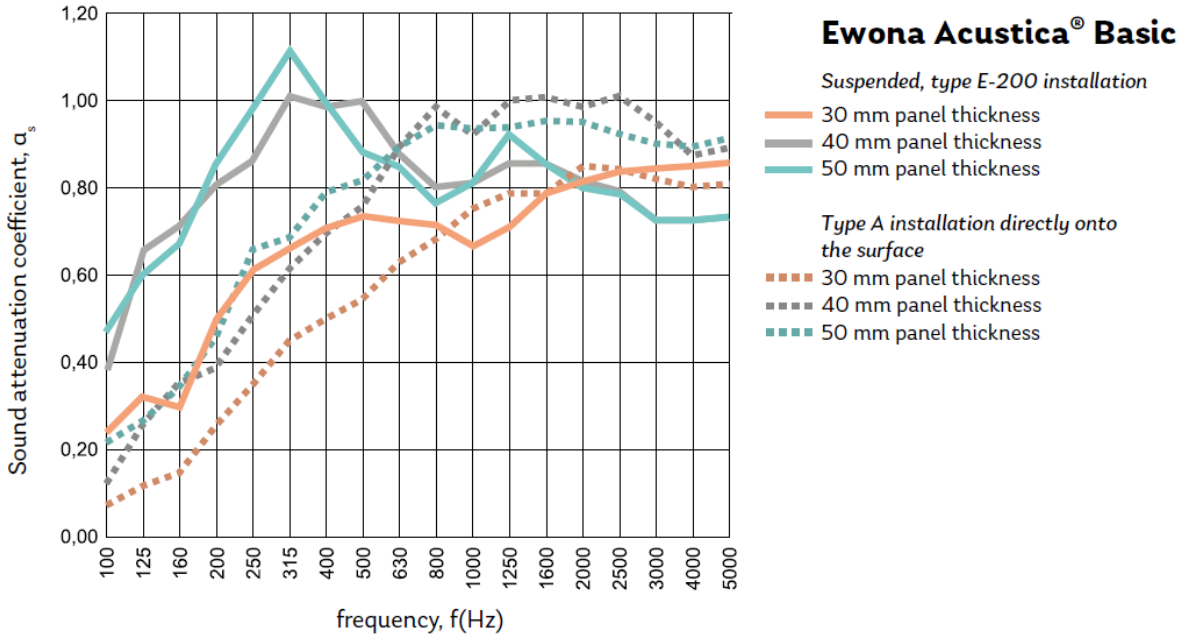


Figure 1: A graph which shows the absorption coefficient as a function of frequency. The panel for Hearing Valley is 50 mm panel with Type A installation. Source: Ewona Acustica Basic datasheet [12].

Boundary Condition for Ceiling

Ceiling panels of the Hearing Valley room are Ecophon Master E T24 acoustic panels. Ecophon provides normal incidence absorption data for acoustics. This implies that impedance BC can also be used in the ceiling. However, this datasheet also excludes low frequencies. Table 3 and Figure 2 shows the practical absorption data provided by Ecophon [13]. The practical absorption coefficient is sufficient approximation of normal incidence absorption coefficient. The data is used for extrapolating for low frequencies. For the ceiling the choice of extrapolation method was piecewise cubic hermite interpolating polynomial (PCHIP). PCHIP was chosen because of the smoothness and the shape preserving properties. The extrapolation was done with MATLAB.

Table 3: Tabular values for α_n provided by Ecophon [13].

Frequencies (Hz)	125	250	500	1000	2000	4000
α_n	0.65	0.85	0.95	1.00	1.00	1.00

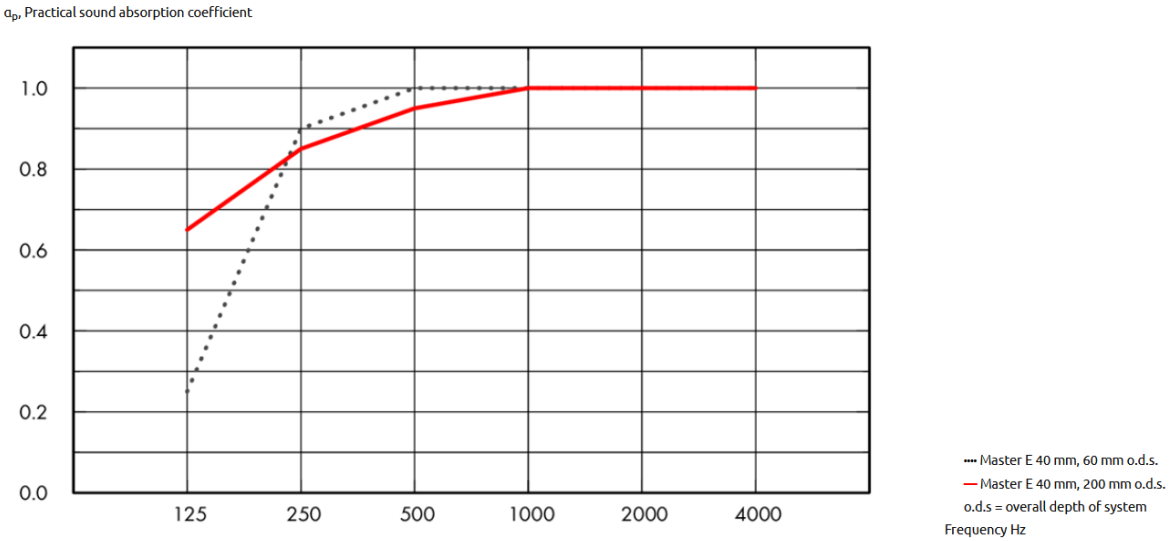


Figure 2: A graph which shows the absorption coefficient as a function of frequency. The panel data used for simulations is Master E 40 mm, 200 mm overall depth of system (o.d.s.) Source: Ecophon [13].

Boundary Condition for Floor

The floor of the Hearing Valley room is carpeting, so the most natural choice for BC is impedance BC (2.14). More detailed information isn't available for this project, and impedance tube was not available. The best guess for normal incidence absorption coefficient was $\alpha_n = 0.3$.

Boundary Conditions for the listener

The listener model is a crude estimate of human model, as one can see from Figure 3. The model is built crudely because of the low frequency acoustics behaves very similarly with or without more accurate details, and crude model is computationally more efficient.

The listener needed a total of 2 different boundary conditions to evaluate. Firstly, the listener has clothing made of fabric on the torso. Impedance BC was chosen for this with $\alpha_n = 0.7$. The absorption coefficient is set a little higher due to the crude estimation of the listener model.

The second BC for the listener is for head and neck. Head and neck are assumed, naturally, as human skin. COMSOL has specific impedance BC setting for human skin. It has impedance defined as

$$z_n = i\omega L + R + \frac{1}{i\omega C}, \tag{3.2}$$

where $R = 6.0 \cdot 10^4 \ \frac{\text{kg}}{\text{m}^2\text{s}}$, $L = 3.53 \ \frac{\text{kg}}{\text{m}^2}$ and $C = 7.95 \cdot 10^{-10} \ \frac{\text{m}^2\text{s}^2}{\text{kg}}$.



Figure 3: The model of the listener.

Boundary Conditions for the speakers

The speaker model consists of the enclosure and the transducer. The enclosure BC was set to Neumann BC (2.6), where the source term $g = 0$. The transducer BC was also set as Neumann BC, but instead of impedance assumption, normal velocity of the boundary is assumed. The source term for (2.6) is derived as $g = i\omega v_n$. where v_n is the normal velocity of the transducer. The normal velocity used in this project was $v_n = 0.1$ m/s.

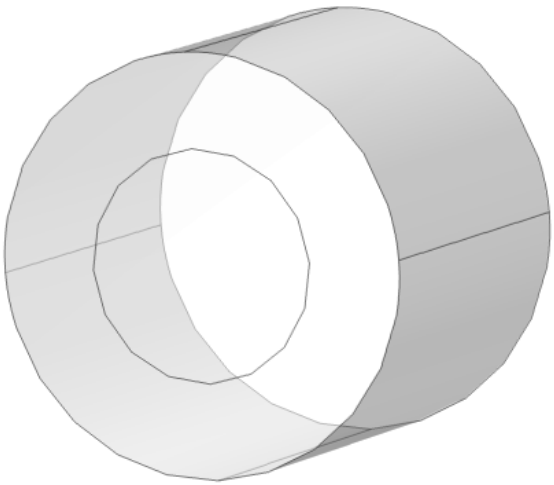


Figure 4: The speaker model used in COMSOL. The transducer diameter is 12 inches, like in the physical version of this speaker model. The physical speaker model is Genelec 7370A [14].

3.1.2 Optimization

Let's define binaural average for two ears, that is

$$m(f) = \frac{x_L(f) + x_R(f)}{2} \tag{3.3}$$

where x_L is the place for left ear and x_R is the place for right ear. Now we can define frequency-weighted mean for continuous frequency case

$$\overline{m} = \frac{\int_{f_{\min}}^{f_{\max}} f\, m(f)\, df}{\int_{f_{\min}}^{f_{\max}} f\, df}. \tag{3.4}$$

Now, by utilizing equations (3.3) and (3.4), we get optimizing problem

$$(m(f) - \overline{m})^2 = \left(\frac{x_L(f) + x_R(f)}{2} - \frac{\int_{f_{\min}}^{f_{\max}} f \frac{x_L(f) + x_R(f)}{2}\, df}{\int_{f_{\min}}^{f_{\max}} f\, df} \right)^2. \tag{3.5}$$

Now discretizing the frequency,

$$(m(f_i) - \overline{m})^2 = \left(\frac{x_L(f_i) + x_R(f_i)}{2} - \frac{\sum_{j=1}^{N} f_j \frac{x_L(f_j) + x_R(f_j)}{2}}{\sum_{j=1}^{N} f_j} \right)^2. \tag{3.6}$$

For this project, the discretization interval of the frequency was set to 1 Hz. The equation (3.6) is the final form for a flat sound optimization problem, which is used in COMSOL. For this optimization problem, we need an optimization method.

COMSOL has multiple different optimization methods. For this project was chosen Nelder-Mead method. Nelder-Mead is a simplex-based method for minimizing multidimensional functions [15]. Simplex methods are gradient-free methods for finding local minima's from functions. Nelder-Mead method expands this to nonlinear functions [15].

Our task is to optimize the location of the listener for the flat frequency response in SPL. This means that our task is to find the minima of the function (3.6) with respect to $m(f_i)$ in the equation (3.3). The height of the listener's ears is fixed in our case, but we can move the listener in *xy*-plane. Nelder-Mead is a good minimizing algorithm for this kind of optimization problem.

3.2 Measurements for Hearing Valley

Validations are important when simulations are done. For this project, measurements were only done to the original place of the listener due to the fact that the listener's place cannot be changed in the physical room. The validation measurements were frequency sweeps from 20 Hz to 150 Hz, which is the frequency response of the speakers [14].

The measurements were done with a single measurement condenser microphone manufactured by Behringer. It is an omnidirectional condenser microphone designed for measurement applications [16].

The measuring software used was Room EQ Wizard (REW), a free room acoustics software that implemented the frequency sweeps within the two subwoofers inside Hearing Valley [17]. The microphone gives the measurement data as acoustic pressure in time-domain. The time-domain acoustic pressure data can be converted into frequency-domain with discrete Fourier-transform,

$$P_k = \sum_{n=0}^{N-1} p_n e^{-i2\pi \frac{k}{N} n}, \tag{3.7}$$

where F_k, $k = 0, 1, 2, ..., N-1$, are the frequency-domain data points, and f_n, $n = 0, 1, 2, ..., N-1$, are the original time-domain data points. Fourier-transform for the whole data set can be expressed in matrix form,

$$P = Wp, \tag{3.8}$$

where W is a matrix with column vectors

$$W = \begin{bmatrix} 1 & e^{-i2\pi \frac{k}{N} \cdot 1} & e^{-i2\pi \frac{k}{N} \cdot 2} & \cdots & e^{-i2\pi \frac{k}{N}(N-1)} \end{bmatrix}.$$

SPL can be calculated by placing frequency response P from (3.8) in the SPL equation (2.15).

REW software does this analysis automatically when the samples are measured. However, it is a good practice to do the analysis, for example, with MATLAB to be sure that the analysis is done correctly for the situation.



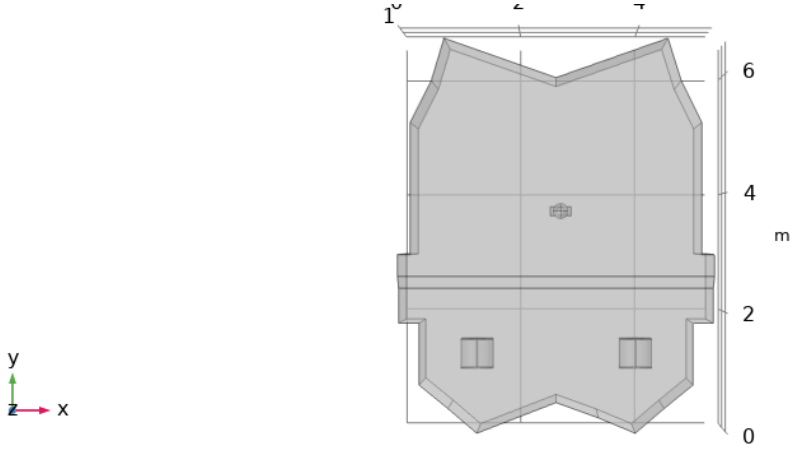
Figure 5: Picture taken from the physical Hearing Valley room. The listener's position is in the middle of the speaker rack.



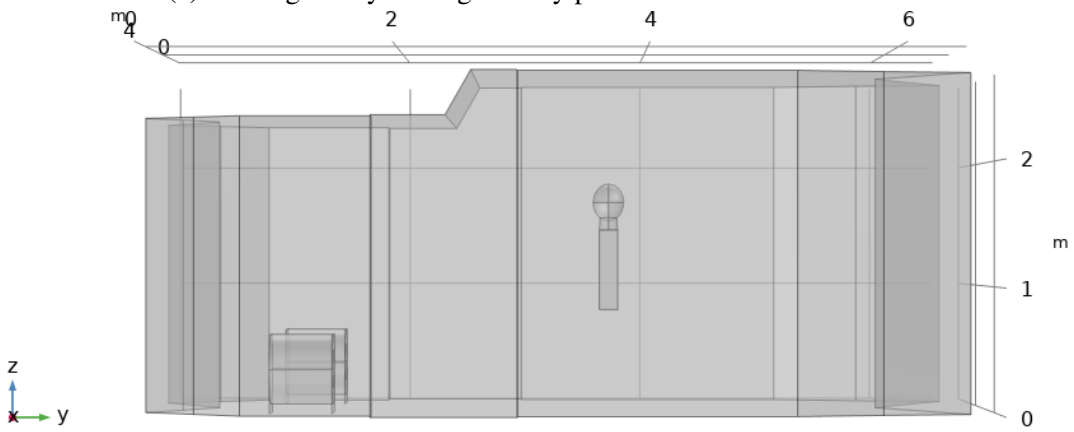
Figure 6: A capture of the physical room. The subwoofers are behind the photographer.



Figure 7: Genelec 7370A subwoofer used in Hearing Valley as a low frequency loudspeaker.



(a) Hearing Valley room geometry presented from above the room.



(b) Hearing Valley room geometry presented from the side of the room.

Figure 8: Hearing Valley room geometry built in COMSOL. The initial position of the listener is presented in these figures.

4 Results

4.1 Simulations

The original location of the listener in the xy-plane was $(270\text{ cm}, 370\text{ cm})$, according to the COMSOL room geometry (Figure 8). As we can see from table 4 and Figure 10, the Nelder-Mead algorithm found at least a local minima from $(209\text{ cm}, 336\text{ cm})$, which is a slight change in the listener's position.

Figure 9 shows the comparison of the SPL in the original and the optimized position for both left and right ear. As we can see, there are some slight differences between the ogirinal and the optimized position. For example, between 30 Hz and 40 Hz, and just below 100 Hz, there is a decline in the original position SPL, which is fixed in optimized position. The SPL was calculated with (2.15).

Table 4: Initial and optimized position for the listener when the goal is as flat sound as possible.

	x-axis	**y-axis**
Initial position	270 cm	370 cm
Optimized position	209 cm	336 cm

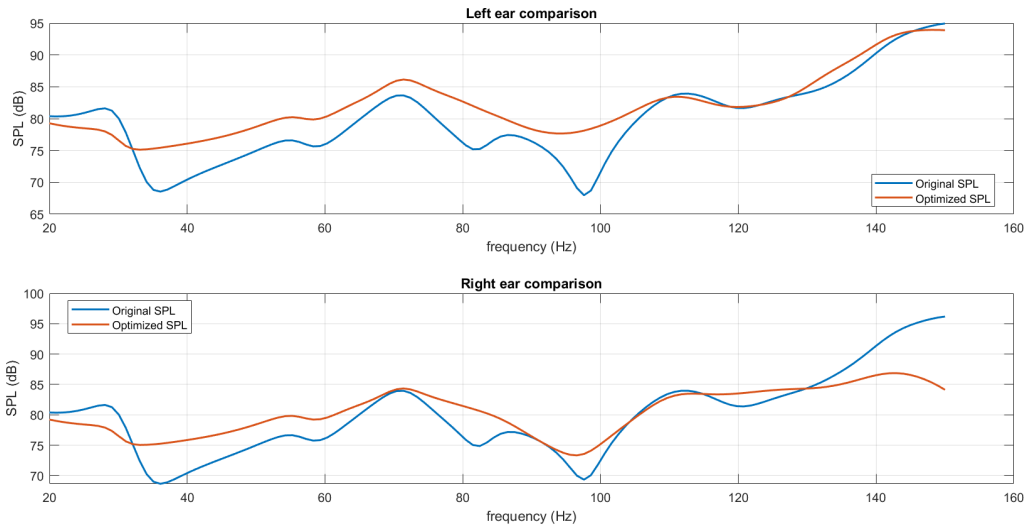
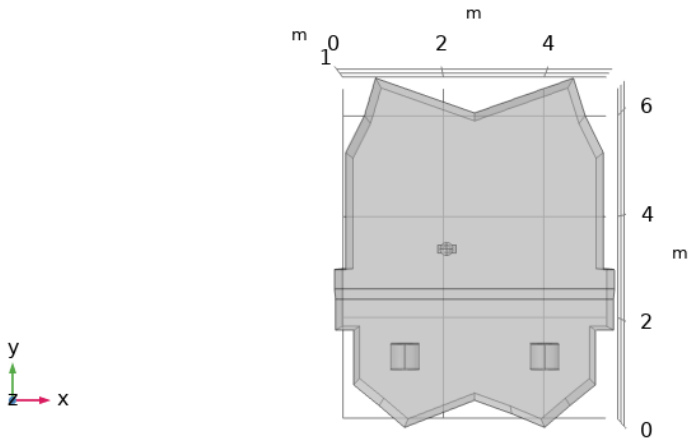
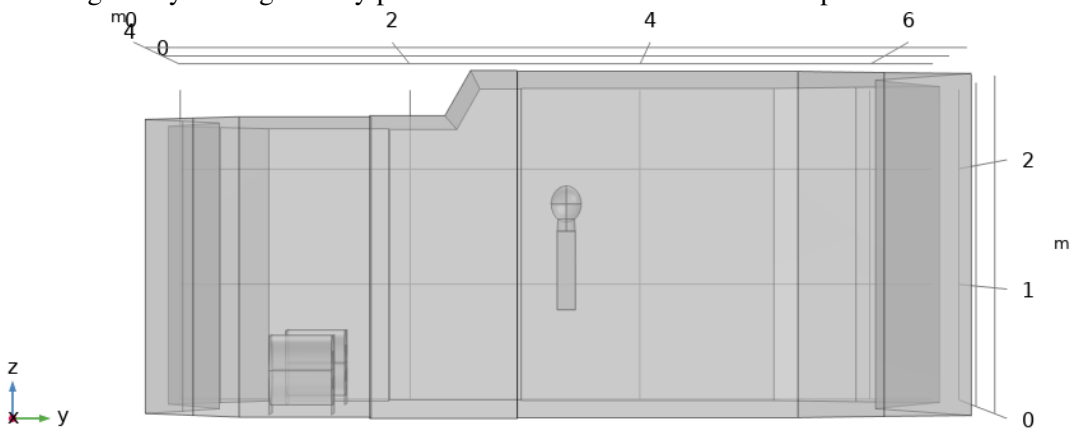


Figure 9: SPL comparison for both ears plotted.



(a) Hearing Valley room geometry presented from above the room with optimized listener's location.



(b) Hearing Valley room geometry presented from the side of the room with optimized listener's location.

Figure 10: Hearing Valley room geometry built in COMSOL. The optimized location of the listener is presented in these figures.

4.2 Comparison of Simulations and Measurements

Validation of the simulations were done by taking the average of left and right ear SPL of the original listener's position and comparing it to real measurements in Hearing Valley. The measurements are not comparable as they are, but the graph curves of the results can be compared.

Figure 11 shows the comparison between the simulation results and the measurements. The graphs have relatively similar shapes. The largest difference between the graphs is the trend. The measured SPL curve has an obvious decreasing trend, and simulated SPL has a relatively flat trend.

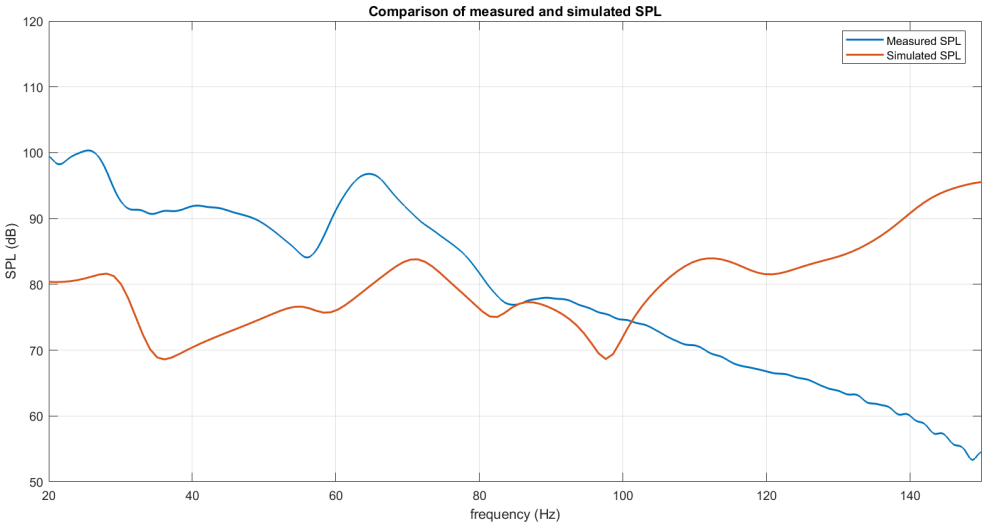


Figure 11: Comparison of the measured SPL and the simulated SPL.

5 Discussion

The results show that frequency response is stable in the vicinity of the original listener's location. That is, small changes done for the listener's location inside the room cause small changes to SPL. The optimization shows a slight change in the listener's location, but as we see from the results, it has only minor effects on the frequency response (Figure 9). The frequency response graph smoothens when the listener's location is at the minimum of the optimization function (Equation (3.6)). Although the changes of the frequency response are relatively small, the comparison graphs show valuable information of the optimization.

Figure 9 shows that below 40 Hz and 100 Hz there are local minimas in the frequency responses of the original listener's location. The local minimas could be a cause of nodes in the acoustic pressure wave. That is, the wave length is either the length of the room or multiple of the length of the room, and causes a node close to the listener's location. The optimization reduces the dip in both of these frequencies.

When we compare the original and the optimized listener's location, we can see that the optimized location is asymmetrical with respect to the subwoofers (Figures 8 and 10). This can be caused by the combined effect of asymmetry of the Hearing Valley room and the impedance boundary conditions of the walls and the ceiling.

The validation measurements show that the simulations are relatively accurate. There are slight changes that can be discussed. The subwoofer positions can only be approximations inside the Hearing Valley room because of how the room geometry is built in COMSOL. The coordinate system can be seen from Figure 8. Because of this inconvenience, the subwoofer position's could be set incorrectly inside the physical room. Furthermore, the impedance BC

is only an approximation for the real boundaries. This approximation could be the main cause for the shift of the simulated SPL when it is compared to the measured SPL in Figure 11.

The subwoofer model is extremely rough approximation of the real Genelec 7370A subwoofer. Building the exact model of the subwoofer would have been too time consuming, and would have caused unnecessary computational ineffectiveness. The simulation results and the measurements can be still compared by the shapes of the frequency response graphs and other similarities.

Figure 11 shows that the measurements of frequency response begins to decrease from 90 Hz to 150 Hz, which is the highest frequency that the subwoofer can produce [14]. The explanation for the decrease in the measurement data comes from the role of the subwoofers in audio systems. They are low frequency balancers that support the rest of the soundscape. That is, subwoofers are not designed to be used alone. In the Hearing Valley audio setup (Figure 5), 3-way loudspeakers are the main part of the audio system, and the subwoofers support the low frequencies from 20 Hz to 100 Hz. The 3-way loudspeaker system can be seen from Figure 5.

The physical Hearing Valley room has curtains installed near the walls, which is not modelled in COMSOL. However, the wavelength of low frequency acoustic pressure waves is relatively large, so the effect of the curtains is minimal to none.

6 Conclusion

The results show that the listener's location is fairly well chosen, and the frequency response is stable in the vicinity of the listener's location. However, if there is a need for flatter frequency response, it can be achieved by optimizing the locations of the subwoofers. This project was a way to see if the listener's location is selected well. The SPL is stable in the vicinity of the listener's original location, and small changes in listener's location doesn't dramatically change the frequency response. This is great information for the physical room because, when studies are done in the room, it doesn't require the listener to be perfectly still, at least for low frequency situations.

It is important to realize that the listener's location is fixed in the physical Hearing Valley room setup. The study of this project was to examine the stability of the frequency response in the vicinity of the fixed listener's location.

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